

Creation Date 16-Jan-2009

Revision Date 02-May-2025

Revision Number 4

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                                  |
|---------------------------|--------------------------------------------------|
| Product Description:      | <b>Methyl formate, 97%, cont. ca 3% methanol</b> |
| Cat No. :                 | <b>L03623</b>                                    |
| Synonyms                  | Formic acid methyl ester; Methyl methanoate      |
| Index No                  | 607-014-00-1                                     |
| CAS No                    | 107-31-3                                         |
| EC No                     | 203-481-7                                        |
| Molecular Formula         | C2 H4 O2                                         |
| REACH registration number | 01-2119433307-44-0306                            |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

#### E-mail address

begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## Section 2: HAZARDS IDENTIFICATION

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## 2.1. Classification of the substance or mixture

### CLP Classification - Regulation (EC) No 1272/2008

#### Physical hazards

Flammable liquids

Category 1 (H224)

#### Health hazards

Acute oral toxicity

Category 4 (H302)

Acute Inhalation Toxicity - Vapors

Category 4 (H332)

Serious Eye Damage/Eye Irritation

Category 2 (H319)

Specific target organ toxicity - (single exposure)

Category 2 (H371)

#### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### **Hazard Statements**

H224 - Extremely flammable liquid and vapor

H319 - Causes serious eye irritation

H335 - May cause respiratory irritation

H302 + H332 - Harmful if swallowed or if inhaled

### **Precautionary Statements**

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P309 + P311 - IF exposed or if you feel unwell: Call a POISON CENTER or doctor/physician

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxic to terrestrial vertebrates

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This product does not contain any known or suspected endocrine disruptors

## Section 3: Composition/information on ingredients

### 3.1. Substances

| Component      | CAS No   | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                            |
|----------------|----------|-------------------|----------|--------------------------------------------------------------------------------------------------------------|
| Methyl formate | 107-31-3 | EEC No. 203-481-7 | >97      | Flam. Liq. 1 (H224)<br>Acute Tox. 4 (H302)<br>Acute Tox. 4 (H332)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335) |
| Methanol       | 67-56-1  | 200-659-6         | <3       | Flam. Liq. 2 (H225)<br>Acute Tox. 3 (H301)<br>Acute Tox. 3 (H311)<br>Acute Tox. 3 (H331)<br>STOT SE 1 (H370) |

| Component | Specific concentration limits (SCL's)                         | M-Factor | Component notes |
|-----------|---------------------------------------------------------------|----------|-----------------|
| Methanol  | STOT Single Exp. 1 :: >= 10<br>STOT Single Exp. 2 :: 3 - < 10 | -        | -               |

|                           |                       |
|---------------------------|-----------------------|
| REACH registration number | 01-2119433307-44-0306 |
|---------------------------|-----------------------|

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                                         |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                      |
| Ingestion                          | Do NOT induce vomiting. Get medical attention.                                                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| Self-Protection of the First Aider | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                 |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

|                    |                                                 |
|--------------------|-------------------------------------------------|
| Notes to Physician | Treat symptomatically. Symptoms may be delayed. |
|--------------------|-------------------------------------------------|

## Section 5: Firefighting measures

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## 5.1. Extinguishing media

### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### **Extinguishing media which must not be used for safety reasons**

No information available.

## 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Extremely flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Methanol.

## 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6: ACCIDENTAL RELEASE MEASURES**

### 6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## **Section 7: Handling and storage**

### 7.1. Precautions for safe handling

Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

### 7.2. Conditions for safe storage, including any incompatibilities

Flammables area. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

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Technical Rules for Hazardous Substances (TRGS) 510  
Storage Class (LGK) (Germany)

Class 3

Switzerland - Storage of hazardous substances

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component      | European Union                                                                                                              | The United Kingdom                                                                                                        | France                                                                                                                                                                                                                                                                                                                                                                                                                                  | Belgium                                                                                                                                | Spain                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl formate | TWA: 125 mg/m <sup>3</sup> (8h)<br>TWA: 50 ppm (8h)<br>STEL: 250 mg/m <sup>3</sup> (15min)<br>STEL: 100 ppm (15min)<br>Skin | STEL: 100 ppm 15 min<br>STEL: 250 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 125 mg/m <sup>3</sup> 8 hr<br>Skin | TWA / VME: 50 ppm (8 heures). indicative limit<br>TWA / VME: 125 mg/m <sup>3</sup> (8 heures). indicative limit<br>STEL / VLCT: 100 ppm. indicative limit<br>STEL / VLCT: 250 mg/m <sup>3</sup> . indicative limit<br>Peau                                                                                                                                                                                                              | TWA: 50 ppm 8 uren<br>TWA: 125 mg/m <sup>3</sup> 8 uren<br>STEL: 100 ppm 15 minuten<br>STEL: 250 mg/m <sup>3</sup> 15 minuten<br>Huid  | STEL / VLA-EC: 100 ppm (15 minutos).<br>STEL / VLA-EC: 250 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 50 ppm (8 horas)<br>TWA / VLA-ED: 125 mg/m <sup>3</sup> (8 horas)<br>Piel |
| Methanol       | TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr<br>Skin                                                                | WEL - TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup> STEL           | TWA / VME: 200 ppm (8 heures). restrictive limit<br>TWA / VME: 260 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 1000 ppm. restrictive limit: this value is not set by regulation and comes from a circular published by the Ministry of Labor.<br>STEL / VLCT: 1300 mg/m <sup>3</sup> . restrictive limit: this value is not set by regulation and comes from a circular published by the Ministry of Labor.<br>Peau | TWA: 200 ppm 8 uren<br>TWA: 266 mg/m <sup>3</sup> 8 uren<br>STEL: 250 ppm 15 minuten<br>STEL: 333 mg/m <sup>3</sup> 15 minuten<br>Huid | TWA / VLA-ED: 200 ppm (8 horas)<br>TWA / VLA-ED: 266 mg/m <sup>3</sup> (8 horas)<br>Piel                                                                                              |

| Component      | Italy                                                                                                                                                                                                 | Germany                                                                                                                                             | Portugal                                                                                                                                | The Netherlands                                                                                                                       | Finland                                                                                                                                              |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl formate | TWA: 125 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>TWA: 50 ppm 8 ore.<br>Time Weighted Average<br>STEL: 250 mg/m <sup>3</sup> 15 minuti. Short-term<br>STEL: 100 ppm 15 minuti. Short-term | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 120 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK | STEL: 100 ppm 15 minutos<br>STEL: 250 mg/m <sup>3</sup> 15 minutos<br>TWA: 50 ppm 8 horas<br>TWA: 125 mg/m <sup>3</sup> 8 horas<br>Pele | huid<br>STEL: 100 ppm 15 minuten<br>STEL: 250 mg/m <sup>3</sup> 15 minuten<br>TWA: 50 ppm 8 uren<br>TWA: 125 mg/m <sup>3</sup> 8 uren | TWA: 50 ppm 8 tunteina<br>TWA: 125 mg/m <sup>3</sup> 8 tunteina<br>STEL: 100 ppm 15 minuutteina<br>STEL: 250 mg/m <sup>3</sup> 15 minuutteina<br>Iho |

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|          |                                                                                                                     |                                                                                                               |                                                                                                      |                                                                  |                                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Pelle                                                                                                               | TWA: 120 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 240 mg/m <sup>3</sup><br>Haut |                                                                                                      |                                                                  |                                                                                                                                                                   |
| Methanol | TWA: 200 ppm 8 ore.<br>Time Weighted Average<br>TWA: 260 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>Pelle | 100 ppm TWA MAK;<br>130 mg/m <sup>3</sup> TWA<br>MAKSkin absorber                                             | STEL: 250 ppm 15<br>minutos<br>TWA: 200 ppm 8 horas<br>TWA: 260 mg/m <sup>3</sup> 8<br>horas<br>Pele | huid<br>TWA: 100 ppm 8 uren<br>TWA: 133 mg/m <sup>3</sup> 8 uren | TWA: 200 ppm 8<br>tunteina<br>TWA: 270 mg/m <sup>3</sup> 8<br>tunteina<br>STEL: 250 ppm 15<br>minuutteina<br>STEL: 330 mg/m <sup>3</sup> 15<br>minuutteina<br>Iho |

| Component      | Austria                                                                                                                                                                                                                     | Denmark                                                                                                                                         | Switzerland                                                                                                                                                   | Poland                                                                                  | Norway                                                                                                                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl formate | Haut<br>MAK-KZGW: 50 ppm 15<br>Minuten<br>MAK-KZGW: 120 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 50 ppm 8<br>Stunden<br>MAK-TMW: 120 mg/m <sup>3</sup><br>8 Stunden<br>Ceiling: 50 ppm<br>Ceiling: 120 mg/m <sup>3</sup> | TWA: 50 ppm 8 timer<br>TWA: 123 mg/m <sup>3</sup> 8 timer<br>STEL: 250 mg/m <sup>3</sup> 15<br>minutter<br>STEL: 100 ppm 15<br>minutter<br>Hud  | Haut/Peau<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 250 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 125 mg/m <sup>3</sup> 8<br>Stunden  | STEL: 200 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 100 mg/m <sup>3</sup> 8<br>godzinach | TWA: 50 ppm 8 timer<br>TWA: 125 mg/m <sup>3</sup> 8 timer<br>STEL: 100 ppm 15<br>minutter. value from the<br>regulation<br>STEL: 250 mg/m <sup>3</sup> 15<br>minutter. value from the<br>regulation<br>Hud |
| Methanol       | Haut<br>MAK-KZGW: 800 ppm<br>15 Minuten<br>MAK-KZGW: 1040<br>mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 200 ppm 8<br>Stunden<br>MAK-TMW: 260 mg/m <sup>3</sup><br>8 Stunden                                                   | TWA: 200 ppm 8 timer<br>TWA: 260 mg/m <sup>3</sup> 8 timer<br>STEL: 400 ppm 15<br>minutter<br>STEL: 520 mg/m <sup>3</sup> 15<br>minutter<br>Hud | Haut/Peau<br>STEL: 400 ppm 15<br>Minuten<br>STEL: 520 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 200 ppm 8<br>Stunden<br>TWA: 260 mg/m <sup>3</sup> 8<br>Stunden | STEL: 300 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 100 mg/m <sup>3</sup> 8<br>godzinach | TWA: 100 ppm 8 timer<br>TWA: 130 mg/m <sup>3</sup> 8 timer<br>STEL: 150 ppm 15<br>minutter. value<br>calculated<br>STEL: 162.5 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated<br>Hud                |

| Component      | Bulgaria                                                                                                     | Croatia                                                                                                                                                                       | Ireland                                                                                                                         | Cyprus                                                                                                                                  | Czech Republic                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Methyl formate | TWA: 125 mg/m <sup>3</sup><br>TWA: 50 ppm<br>STEL : 250 mg/m <sup>3</sup><br>STEL : 100 ppm<br>Skin notation | kože<br>TWA-GVI: 50 ppm 8<br>satima.<br>TWA-GVI: 125 mg/m <sup>3</sup> 8<br>satima.<br>STEL-KGVI: 100 ppm<br>15 minutama.<br>STEL-KGVI: 250 mg/m <sup>3</sup><br>15 minutama. | TWA: 50 ppm 8 hr.<br>TWA: 125 mg/m <sup>3</sup> 8 hr.<br>STEL: 250 mg/m <sup>3</sup> 15<br>min<br>STEL: 100 ppm 15 min<br>Skin  | Skin-potential for<br>cutaneous absorption<br>STEL: 250 mg/m <sup>3</sup><br>STEL: 100 ppm<br>TWA: 125 mg/m <sup>3</sup><br>TWA: 60 ppm | TWA: 125 mg/m <sup>3</sup> 8<br>hodinách.<br>Ceiling: 250 mg/m <sup>3</sup>                                           |
| Methanol       | TWA: 200 ppm<br>TWA: 260.0 mg/m <sup>3</sup><br>Skin notation                                                | kože<br>TWA-GVI: 200 ppm 8<br>satima.<br>TWA-GVI: 260 mg/m <sup>3</sup> 8<br>satima.                                                                                          | TWA: 200 ppm 8 hr.<br>TWA: 260 mg/m <sup>3</sup> 8 hr.<br>STEL: 600 ppm 15 min<br>STEL: 780 mg/m <sup>3</sup> 15<br>min<br>Skin | Skin-potential for<br>cutaneous absorption<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                | TWA: 250 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 1000 mg/m <sup>3</sup> |

| Component      | Estonia                                                                                                                                                           | Gibraltar                                                                                                                             | Greece                                                                                                                                    | Hungary                                                                                                                                                                                                         | Iceland                                                                                                                                            |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl formate | Nahk<br>TWA: 125 mg/m <sup>3</sup> 8<br>tundides.<br>TWA: 50 ppm 8<br>tundides.<br>STEL: 100 ppm 15<br>minutites.<br>STEL: 250 mg/m <sup>3</sup> 15<br>minutites. | Skin notation<br>TWA: 125 mg/m <sup>3</sup> 8 hr<br>TWA: 50 ppm 8 hr<br>STEL: 250 mg/m <sup>3</sup> 15<br>min<br>STEL: 100 ppm 15 min | skin - potential for<br>cutaneous absorption<br>STEL: 100 ppm<br>STEL: 250 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 125 mg/m <sup>3</sup> | STEL: 250 mg/m <sup>3</sup> 15<br>percekben. CK<br>STEL: 100 ppm 15<br>percekben. CK<br>TWA: 125 mg/m <sup>3</sup> 8<br>órában. AK<br>TWA: 50 ppm 8 órában.<br>AK<br>lehetséges borön<br>keresztüli felszívódás | STEL: 100 ppm<br>STEL: 250 mg/m <sup>3</sup><br>TWA: 50 ppm 8<br>klukkustundum.<br>TWA: 125 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Skin notation |
| Methanol       | Nahk<br>TWA: 200 ppm 8<br>tundides.<br>TWA: 250 mg/m <sup>3</sup> 8                                                                                               | Skin notation<br>TWA: 200 ppm 8 hr<br>TWA: 260 mg/m <sup>3</sup> 8 hr                                                                 | skin - potential for<br>cutaneous absorption<br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup>                                              | TWA: 260 mg/m <sup>3</sup> 8<br>órában. AK<br>TWA: 200 ppm 8<br>órában. AK                                                                                                                                      | TWA: 200 ppm 8<br>klukkustundum.<br>TWA: 260 mg/m <sup>3</sup> 8<br>klukkustundum.                                                                 |

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|  |                                                                                             |  |                                            |                                            |                                                                     |
|--|---------------------------------------------------------------------------------------------|--|--------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|
|  | tundides.<br>STEL: 250 ppm 15<br>minutites.<br>STEL: 350 mg/m <sup>3</sup> 15<br>minutites. |  | TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | lehetséges borön<br>keresztüli felszívódás | Skin notation<br>Ceiling: 400 ppm<br>Ceiling: 520 mg/m <sup>3</sup> |
|--|---------------------------------------------------------------------------------------------|--|--------------------------------------------|--------------------------------------------|---------------------------------------------------------------------|

| Component      | Latvia                                                                                                                                  | Lithuania                                                                                                  | Luxembourg                                                                                                                                                                                               | Malta                                                                                                                                                                        | Romania                                                                                                                                          |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl formate | skin - potential for<br>cutaneous exposure<br>STEL: 250 mg/m <sup>3</sup><br>STEL: 100 ppm<br>TWA: 125 mg/m <sup>3</sup><br>TWA: 50 ppm | TWA: 125 mg/m <sup>3</sup> IPRD<br>TWA: 50 ppm IPRD<br>Oda<br>STEL: 250 mg/m <sup>3</sup><br>STEL: 100 ppm | Possibility of significant<br>uptake through the skin<br>TWA: 125 mg/m <sup>3</sup> 8<br>Stunden<br>TWA: 50 ppm 8<br>Stunden<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 250 mg/m <sup>3</sup> 15<br>Minuten | possibility of significant<br>uptake through the skin<br>TWA: 50 ppm<br>TWA: 125 mg/m <sup>3</sup><br>STEL: 100 ppm 15<br>minuti<br>STEL: 250 mg/m <sup>3</sup> 15<br>minuti | Skin notation<br>TWA: 50 ppm 8 ore<br>TWA: 125 mg/m <sup>3</sup> 8 ore<br>STEL: 100 ppm 15<br>minute<br>STEL: 250 mg/m <sup>3</sup> 15<br>minute |
| Methanol       | skin - potential for<br>cutaneous exposure<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                | TWA: 200 ppm IPRD<br>TWA: 260 mg/m <sup>3</sup> IPRD<br>Oda                                                | Possibility of significant<br>uptake through the skin<br>TWA: 200 ppm 8<br>Stunden<br>TWA: 260 mg/m <sup>3</sup> 8<br>Stunden                                                                            | possibility of significant<br>uptake through the skin<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                                                          | Skin notation<br>TWA: 200 ppm 8 ore<br>TWA: 260 mg/m <sup>3</sup> 8 ore                                                                          |

| Component      | Russia                                                                      | Slovak Republic                                                                                                      | Slovenia                                                                                                                                      | Sweden                                                                                                                                                                                | Turkey                                                           |
|----------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Methyl formate |                                                                             | Ceiling: 250 mg/m <sup>3</sup><br>Potential for cutaneous<br>absorption<br>TWA: 125 mg/m <sup>3</sup><br>TWA: 50 ppm | TWA: 50 ppm 8 urah<br>TWA: 125 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 100 ppm 15<br>minutah<br>STEL: 250 mg/m <sup>3</sup> 15<br>minutah   | Binding STEL: 100 ppm<br>15 minuter<br>Binding STEL: 250<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 50 ppm 8 timmar.<br>NGV<br>TLV: 125 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud        |                                                                  |
| Methanol       | TWA: 5 mg/m <sup>3</sup> 1250<br>Skin notation<br>MAC: 15 mg/m <sup>3</sup> | Potential for cutaneous<br>absorption<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup>                                  | TWA: 200 ppm 8 urah<br>TWA: 260 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 800 ppm 15<br>minutah<br>STEL: 1040 mg/m <sup>3</sup> 15<br>minutah | Indicative STEL: 250<br>ppm 15 minuter<br>Indicative STEL: 350<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 200 ppm 8 timmar.<br>NGV<br>TLV: 250 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud | Deri<br>TWA: 200 ppm 8 saat<br>TWA: 260 mg/m <sup>3</sup> 8 saat |

## Biological limit values

List source(s):

| Component | European Union | United Kingdom | France                          | Spain                                   | Germany                                                                                                                                                    |
|-----------|----------------|----------------|---------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methanol  |                |                | Methanol: urine end of<br>shift | Methanol: 15 mg/L urine<br>end of shift | Methanol: 15 mg/L urine<br>(end of shift )<br>Methanol: 15 mg/L urine<br>(for long-term<br>exposures: at the end of<br>the shift after several<br>shifts ) |

| Component | Italy | Finland | Denmark | Bulgaria | Romania                                |
|-----------|-------|---------|---------|----------|----------------------------------------|
| Methanol  |       |         |         |          | Methanol: 6 mg/L urine<br>end of shift |

| Component | Gibraltar | Latvia | Slovak Republic                                                                                                                           | Luxembourg | Turkey |
|-----------|-----------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Methanol  |           |        | Methanol: 30 mg/L urine<br>end of exposure or work<br>shift<br>Methanol: 30 mg/L urine<br>after all work shifts for<br>long-term exposure |            |        |

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## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers; See table for values

| Component                          | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Methyl formate<br>107-31-3 ( >97 ) |                              |                                 |                                | DNEL = 17.1mg/kg<br>bw/day        |
| Methanol<br>67-56-1 ( <3 )         |                              | DNEL = 20mg/kg<br>bw/day        |                                | DNEL = 20mg/kg<br>bw/day          |

| Component                          | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Methyl formate<br>107-31-3 ( >97 ) |                                  |                                     | DNEL = 120mg/m <sup>3</sup>        | DNEL = 120mg/m <sup>3</sup>           |
| Methanol<br>67-56-1 ( <3 )         | DNEL = 130mg/m <sup>3</sup>      | DNEL = 130mg/m <sup>3</sup>         | DNEL = 130mg/m <sup>3</sup>        | DNEL = 130mg/m <sup>3</sup>           |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                          | Fresh water      | Fresh water sediment             | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)            |
|------------------------------------|------------------|----------------------------------|--------------------|------------------------------------|-------------------------------|
| Methyl formate<br>107-31-3 ( >97 ) | PNEC = 0.115mg/L | PNEC = 0.439mg/kg<br>sediment dw | PNEC = 1.15mg/L    | PNEC = 8117mg/L                    | PNEC = 0.0202mg/kg soil<br>dw |
| Methanol<br>67-56-1 ( <3 )         | PNEC = 20.8mg/L  | PNEC = 77mg/kg<br>sediment dw    | PNEC = 1540mg/L    | PNEC = 100mg/L                     | PNEC = 100mg/kg<br>soil dw    |

| Component                          | Marine water      | Marine water sediment             | Marine water Intermittent | Food chain | Air |
|------------------------------------|-------------------|-----------------------------------|---------------------------|------------|-----|
| Methyl formate<br>107-31-3 ( >97 ) | PNEC = 0.0115mg/L | PNEC = 0.0439mg/kg<br>sediment dw |                           |            |     |
| Methanol<br>67-56-1 ( <3 )         | PNEC = 2.08mg/L   | PNEC = 7.7mg/kg<br>sediment dw    |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment



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**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material            | Breakthrough time | Glove thickness | EU standard | Glove comments                                                                 |
|---------------------------|-------------------|-----------------|-------------|--------------------------------------------------------------------------------|
| Butyl rubber<br>Viton (R) | > 30 minutes      | 0.7 mm          | Level 2     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                                                |                                             |                                          |
|------------------------------------------------|---------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                      |                                          |
| <b>Appearance</b>                              | Colorless                                   |                                          |
| <b>Odor</b>                                    | Organic                                     |                                          |
| <b>Odor Threshold</b>                          | No data available                           |                                          |
| <b>Melting Point/Range</b>                     | -100 °C / -148 °F                           |                                          |
| <b>Softening Point</b>                         | No data available                           |                                          |
| <b>Boiling Point/Range</b>                     | 31 - 33 °C / 88 - 91 °F                     |                                          |
| <b>Flammability (liquid)</b>                   | Extremely flammable                         | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                              | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 5 vol%<br><b>Upper</b> 23 vol% |                                          |
| <b>Flash Point</b>                             | -32 °C / -25 °F                             | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 440 °C / 824 °F                             |                                          |
| <b>Decomposition Temperature</b>               | No data available                           |                                          |
| <b>pH</b>                                      | 4-5 @ 20°C                                  | (20 %)                                   |
| <b>Viscosity</b>                               | 0.35 mPa.s at 20 °C                         |                                          |
| <b>Water Solubility</b>                        | 300 g/l (20°C)                              |                                          |
| <b>Solubility in other solvents</b>            | No information available                    |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                             |                                          |
| <b>Component</b>                               | <b>log Pow</b>                              |                                          |
| Methyl formate                                 | -0.21                                       |                                          |
| Methanol                                       | -0.74                                       |                                          |

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|                            |                         |             |
|----------------------------|-------------------------|-------------|
| Vapor Pressure             | 644 mbar @ 20 °C        |             |
| Density / Specific Gravity | 0.968                   |             |
| Bulk Density               | Not applicable          | Liquid      |
| Vapor Density              | 2.07                    | (Air = 1.0) |
| Particle characteristics   | Not applicable (liquid) |             |

## 9.2. Other information

|                      |                                             |
|----------------------|---------------------------------------------|
| Molecular Formula    | C2 H4 O2                                    |
| Molecular Weight     | 60.05                                       |
| Explosive Properties | Vapors may form explosive mixtures with air |

## Section 10: Stability and reactivity

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. . Bases. Acids. Incompatible with oxidizing agents.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Methanol.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Category 4                                                       |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Category 4                                                       |

| Component      | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|----------------|--------------------------------|-------------------------------|-------------------------------|
| Methyl formate | LD50 = 475 mg/kg ( Rat )       | LD50 > 5 g/kg ( Rabbit )      | LC50 > 21 mg/L ( Rat ) 4 h    |
| Methanol       | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

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**(d) respiratory or skin sensitization;**

**Respiratory**  
**Skin**

Based on available data, the classification criteria are not met  
Based on available data, the classification criteria are not met

| Component                  | Test method                                                       | Test species | Study result    |
|----------------------------|-------------------------------------------------------------------|--------------|-----------------|
| Methanol<br>67-56-1 ( <3 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | non-sensitising |

**(e) germ cell mutagenicity;**

Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

Based on available data, the classification criteria are not met

| Component                  | Test method             | Test species / Duration          | Study result              |
|----------------------------|-------------------------|----------------------------------|---------------------------|
| Methanol<br>67-56-1 ( <3 ) | OECD Test Guideline 416 | Rat / Inhalation<br>2 Generation | NOAEC =<br>1.3 mg/l (air) |

**(h) STOT-single exposure;**

Category 2

**Results / Target organs**

Respiratory system, Optic nerve, Central nervous system (CNS).

**(i) STOT-repeated exposure;**

Based on available data, the classification criteria are not met

**Target Organs**

None known.

**(j) aspiration hazard;**

Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed**

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

**Ecotoxicity effects**

Do not empty into drains.

| Component      | Freshwater Fish                               | Water Flea                               | Freshwater Algae                                                                                         |
|----------------|-----------------------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Methyl formate |                                               | EC50: > 500 mg/L, 48h<br>(Daphnia magna) | EC50: = 240 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: = 190 mg/L, 96h<br>(Desmodesmus subspicatus) |
| Methanol       | Pimephales promelas: LC50 ><br>10000 mg/L 96h | EC50 > 10000 mg/L 24h                    |                                                                                                          |

| Component      | Microtox                 | M-Factor |
|----------------|--------------------------|----------|
| Methyl formate | EC50 > 10000 mg/L 17 h   |          |
| Methanol       | EC50 = 39000 mg/L 25 min |          |

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|  |                                                     |  |
|--|-----------------------------------------------------|--|
|  | EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min |  |
|--|-----------------------------------------------------|--|

## **12.2. Persistence and degradability**

### **Persistence**

Readily biodegradable  
Persistence is unlikely, based on information available.

| Component                | Degradability                  |
|--------------------------|--------------------------------|
| Methanol<br>67-56-1 (<3) | DT50 ~ 17.2d<br>>94% after 20d |

## **12.3. Bioaccumulative potential**

Bioaccumulation is unlikely

| Component      | log Pow | Bioconcentration factor (BCF) |
|----------------|---------|-------------------------------|
| Methyl formate | -0.21   | No data available             |
| Methanol       | -0.74   | <10 dimensionless             |

## **12.4. Mobility in soil**

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

## **12.5. Results of PBT and vPvB assessment**

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## **12.6. Endocrine disrupting properties**

### **Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

## **12.7. Other adverse effects**

### **Persistent Organic Pollutant Ozone Depletion Potential**

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## **Section 13: Disposal considerations**

### **13.1. Waste treatment methods**

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### **European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

#### **Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## **Section 14: Transport information**

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## IMDG/IMO

**14.1. UN number** UN1243  
**14.2. UN proper shipping name** Methyl formate  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** I

## ADR

**14.1. UN number** UN1243  
**14.2. UN proper shipping name** Methyl formate  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** I

## IATA

**14.1. UN number** UN1243  
**14.2. UN proper shipping name** Methyl formate  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** I

**14.5. Environmental hazards** No hazards identified  
**14.6. Special precautions for user** No special precautions required.  
**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component      | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|----------------|----------|-----------|--------|-----|-------|------|----------|------|------|
| Methyl formate | 107-31-3 | 203-481-7 | -      | -   | X     | X    | KE-17243 | X    | X    |
| Methanol       | 67-56-1  | 200-659-6 | -      | -   | X     | X    | KE-23193 | X    | X    |

| Component      | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|----------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Methyl formate | 107-31-3 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Methanol       | 67-56-1  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component      | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl formate | 107-31-3 | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)           | -                                                                                                     |

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|          |         |   |                                                                                                                                            |   |
|----------|---------|---|--------------------------------------------------------------------------------------------------------------------------------------------|---|
| Methanol | 67-56-1 | - | Use restricted. See entry 69.<br>(see link for restriction details)<br>Use restricted. See entry 75.<br>(see link for restriction details) | - |
|----------|---------|---|--------------------------------------------------------------------------------------------------------------------------------------------|---|

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component      | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|----------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyl formate | 107-31-3 | Not applicable                                                                            | Not applicable                                                                           |
| Methanol       | 67-56-1  | 500 tonne                                                                                 | 5000 tonne                                                                               |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component      | Germany - Water Classification (AwSV) | Germany - TA-Luft Class                    |
|----------------|---------------------------------------|--------------------------------------------|
| Methyl formate | WGK1<br>WGK2                          | Class II : 0.10 g/m³ (Massenkonzentration) |
| Methanol       | WGK 2                                 | Class I : 20 mg/m³ (Massenkonzentration)   |

| Component      | France - INRS (Tables of occupational diseases)      |
|----------------|------------------------------------------------------|
| Methyl formate | Tableaux des maladies professionnelles (TMP) - RG 84 |
| Methanol       | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                          | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Methyl formate<br>107-31-3 ( >97 ) |                                                                                                                | Group I                                                                         |                                                                                             |
| Methanol<br>67-56-1 ( <3 )         | Prohibited and Restricted Substances                                                                           | Group I                                                                         |                                                                                             |

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## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H224 - Extremely flammable liquid and vapor

H225 - Highly flammable liquid and vapor

H302 - Harmful if swallowed

H319 - Causes serious eye irritation

H370 - Causes damage to organs

H335 - May cause respiratory irritation

H301 - Toxic if swallowed

H311 - Toxic in contact with skin

H331 - Toxic if inhaled

H332 - Harmful if inhaled

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

### Prepared By

Health, Safety and Environmental Department

### Creation Date

16-Jan-2009

### Revision Date

02-May-2025

### Revision Summary

Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.**

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## COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**